Администрирование сетевых подсистем

Лабораторная работа № 9 — Настройка POP3/IMAP сервера (Postfix + Dovecot)

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Цели и задачи работы

Цель лабораторной работы

Приобрести практические навыки установки и простейшего конфигурирования POP3/IMAP-сервера.

Реализовать доступ к почте по IMAP/POP3 и проверить работу сервиса с клиентской стороны.

Выполнение лабораторной работы

Разрешённые протоколы IMAP/POP3

Open | dovecot.conf /etc/dovecot | Save

```
7
8 # '#' character and everything after it is treated as comments. Extra spaces
9 # and tabs are ignored. If you want to use either of these explicitly, put the
10 # value inside quotes, eg.: key = "# char and trailing whitespace  "
11
12 # Most (but not all) settings can be overridden by different protocols and/or
13 # source/destination IPs by placing the settings inside sections, for example:
14 # protocol imap { }, local 127.0.0.1 { }, remote 10.0.0.0/8 { }
15
16 # Default values are shown for each setting, it's not required to uncomment
17 # those. These are exceptions to this though: No sections (e.g. namespace {})
18 # or plugin settings are added by default, they're listed only as examples.
19 # Paths are also just examples with the real defaults being based on configure
20 # options. The paths listed here are for configure --prefix=/usr
21 # --sysconfdir=/etc --localstatedir=/var
22
23 # Protocols we want to be serving.
24
25 protocols = imap pop3
26
27 # A comma separated list of IPs or hosts where to listen in for connections.
28 # "*" listens in all IPv4 interfaces, "::" listens in all IPv6 interfaces.
29 # If you want to specify non-default ports or anything more complex,
30 # edit conf.d/master.conf.
31 #listen = *, ::
```


Рис. 1: Настройка списка протоколов в dovecot.conf

Проверка механизма аутентификации

Open | 10-auth.conf | /etc/dovecot/conf.d | Save

```
88 # Require a valid SSL client certificate or the authentication fails.
89 #auth_ssl_require_client_cert = no
90
91 # Take the username from client's SSL certificate, using
92 # X509_NAME_get_text_by_NID() which returns the subject's DN's
93 # CommonName.
94 #auth_ssl_username_from_cert = no
95
96 # Space separated list of wanted authentication mechanisms:
97 #   plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp
98 #   gss-spnego
99 # NOTE: See also disable_plaintext_auth setting.
100 auth_mechanisms = plain
101
102 ##
103 ## Password and user databases
104 ##
105
106 #
107 # Password database is used to verify user's password (and nothing more).
108 # You can have multiple passdbs and userdbs. This is useful if you want to
109 # allow both system users (/etc/passwd) and virtual users to login without
110 # duplicating the system users into virtual database.
111 #
112 # <doc/wiki/PasswordDatabase.txt>
```

Источники пользователей и паролей (PAM + passwd)

Open

auth-system.conf.ext
/etc/dovecot/conf.d

```
1 # Authentication for system users. Included from 10-auth.conf.
2 #
3 # <doc/wiki/PasswordDatabase.txt>
4 # <doc/wiki/UserDatabase.txt>
5
6 # PAM authentication. Preferred nowadays by most systems.
7 # PAM is typically used with either userdb passwd or userdb static.
8 # REMEMBER: You'll need /etc/pam.d/dovecot file created for PAM
9 # authentication to actually work. <doc/wiki/PasswordDatabase.PAM.txt>
10 passdb {
11   driver = pam
12 }
13
14 userdb {
15   driver = passwd
16 }
17
18 # Static settings generated from template <doc/wiki/UserDatabase.Static.txt>
19 #userdb {
20   #driver = static
21   # Can return anything a userdb could normally return. For example:
22   #
23   #  args = uid=500 gid=500 home=/var/mail/%u
24   #
25   # LDA and LMTP needs to look up users only from the userdb. This of course
```

Расположение почтовых ящиков (Maildir)

Open | 10-mail.conf /etc/dovecot/conf.d | Save

```
16 #
17 #   %u - username
18 #   %n - user part in user@domain, same as %u if there's no domain
19 #   %d - domain part in user@domain, empty if there's no domain
20 #   %h - home directory
21 #
22 # See doc/wiki/Variables.txt for full list. Some examples:
23 #
24 #   mail_location = maildir:~/Maildir
25 #   mail_location = mbox:~/mail:INBOX=/var/mail/%u
26 #   mail_location = mbox:/var/mail/%d/%1n/%n:INDEX=/var/indexes/%d/%1n/%n
27 #
28 # <doc/wiki/MailLocation.txt>
29 #
30
31 mail_location = maildir:~/Maildir
32
33 # If you need to set multiple mailbox locations or want to change default
34 # namespace settings, you can do it by defining namespace sections.
35 #
36 # You can have private, shared and public namespaces. Private namespaces
37 # are for user's personal mails. Shared namespaces are for accessing other
38 # users' mailboxes that have been shared. Public namespaces are for shared
39 # mailboxes that are managed by sysadmin. If you create any shared or public
40 # namespaces you'll typically want to enable ACL plugin also, otherwise all
```

Межсетевой экран и запуск служб

```
[root@server.akabrikosov.net server]#
[root@server.akabrikosov.net server]# postconf -e 'home_mailbox = Maildir/'
[root@server.akabrikosov.net server]# firewall-cmd --add-service=pop3 --permanent
success
[root@server.akabrikosov.net server]# firewall-cmd --add-service=pop3s --permanent
success
[root@server.akabrikosov.net server]# firewall-cmd --add-service=imap --permanent
success
[root@server.akabrikosov.net server]# firewall-cmd --add-service=imaps --permanent
success
[root@server.akabrikosov.net server]# firewall-cmd --reload
success
[root@server.akabrikosov.net server]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps pop3 pop3s smtp ssh ssh-custom
[root@server.akabrikosov.net server]# restorecon -vR /etc
[root@server.akabrikosov.net server]# systemctl restart postfix.service
[root@server.akabrikosov.net server]# systemctl enable dovecot
Created symlink '/etc/systemd/system/multi-user.target.wants/dovecot.service' → '/usr/lib/systemd/sys
tem/dovecot.service'.
[root@server.akabrikosov.net server]# systemctl start dovecot
[root@server.akabrikosov.net server]#
```

Рис. 5: Настройка межсетевого экрана для почтовых сервисов

Мониторинг работы почтовых служб

```
Jan  8 09:36:46 server postfix/qmgr[20279]: C0E14413ACAF: from=<>, size=2624, nrcpt=1 (queue active)
Jan  8 09:36:46 server postfix/smtp[21059]: connect to client.akabrikosov.net[192.168.1.30]:25: No ro
ute to host
Jan  8 09:36:46 server postfix/smtp[21059]: C0E14413ACAF: to=<akabrikosov@client.akabrikosov.net>, re
lay=none, delay=2628, delays=2628/0.02/0/0, dsn=4.4.1, status=deferred (connect to client.akabrikosov
.net[192.168.1.30]:25: No route to host)
Jan  8 09:38:32 server dovecot[20490]: imap-login: Login: user=<akabrikosov>, method=PLAIN, rip=192.1
68.1.30, lip=192.168.1.1, mpid=21285, TLS, session=<ENYGLN1HtujAqAEe>
Jan  8 09:38:55 server postfix/smtpd[21339]: connect from client.akabrikosov.net[192.168.1.30]
Jan  8 09:38:55 server postfix/smtpd[21339]: 6A7BA412AD61: client=client.akabrikosov.net[192.168.1.30
]
Jan  8 09:38:55 server postfix/cleanup[21342]: 6A7BA412AD61: message-id=<4410500495ef26c431d47d8cfe57
b95ae53b4d24.camel@akabrikosov.net>
Jan  8 09:38:55 server postfix/qmgr[20279]: 6A7BA412AD61: from=<akabrikosov@akabrikosov.net>, size=57
0, nrcpt=1 (queue active)
Jan  8 09:38:55 server postfix/smtpd[21339]: disconnect from client.akabrikosov.net[192.168.1.30] ehl
o=1 mail=1 rcpt=1 data=1 quit=1 commands=5
Jan  8 09:38:55 server postfix/local[21343]: 6A7BA412AD61: to=<akabrikosov@akabrikosov.net>, relay=lo
cal, delay=0.01, delays=0.01/0/0/0, dsn=2.0.0, status=sent (delivered to maildir)
Jan  8 09:38:55 server postfix/qmgr[20279]: 6A7BA412AD61: removed
```

Рис. 6: Записи в журнале Postfix и Dovecot

Проверка почтового ящика на сервере

```
[akabrikosov@server.akabrikosov.net ~]$
[akabrikosov@server.akabrikosov.net ~]$ MAIL=~/Maildir/ mail
s-nail version v14.9.24.  Type `?' for help
/home/akabrikosov/Maildir: 3 messages 3 unread
►U  1 akabrikosov           2026-01-08 09:38   18/685   "test1
 U  2 akabrikosov           2026-01-08 09:39   18/685   "test2
 U  3 akabrikosov           2026-01-08 09:40   18/685   "test3
& q
Held 3 messages in /home/akabrikosov/Maildir
[akabrikosov@server.akabrikosov.net ~]$
```

Рис. 7: Просмотр почтового ящика пользователя на сервере

Evolution: идентификация пользователя

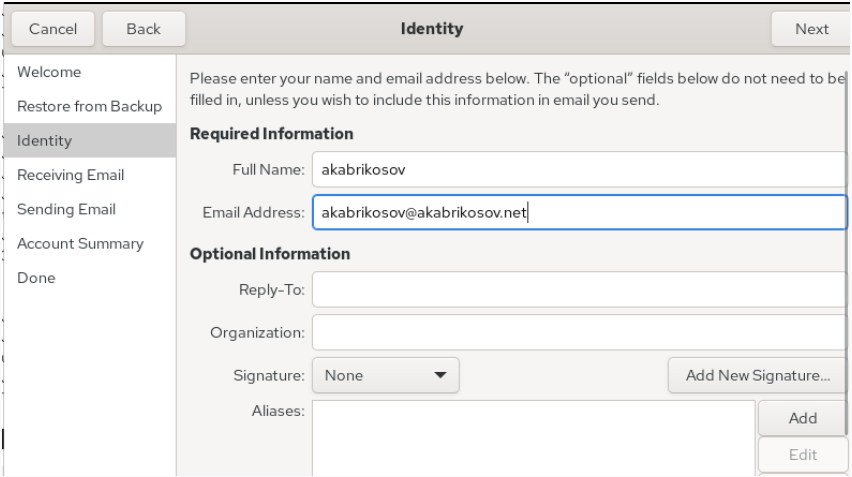


Рис. 8: Настройка учетной записи в Evolution — идентификация

Evolution: входящая почта (IMAP)

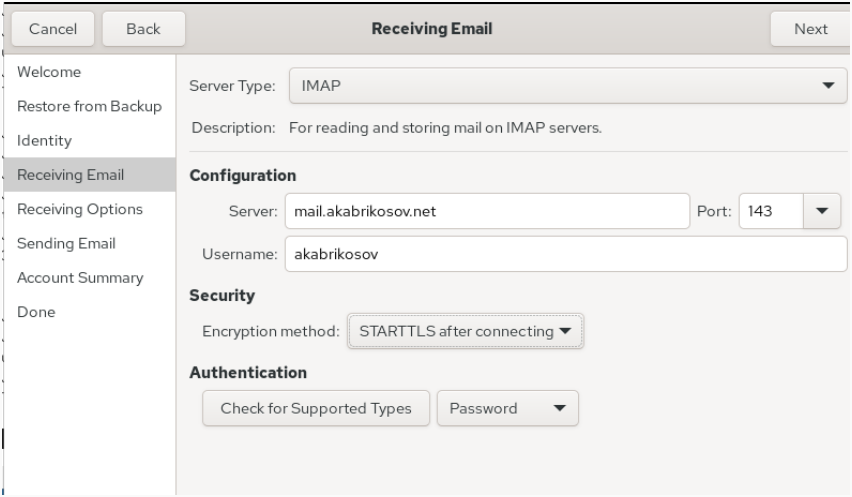
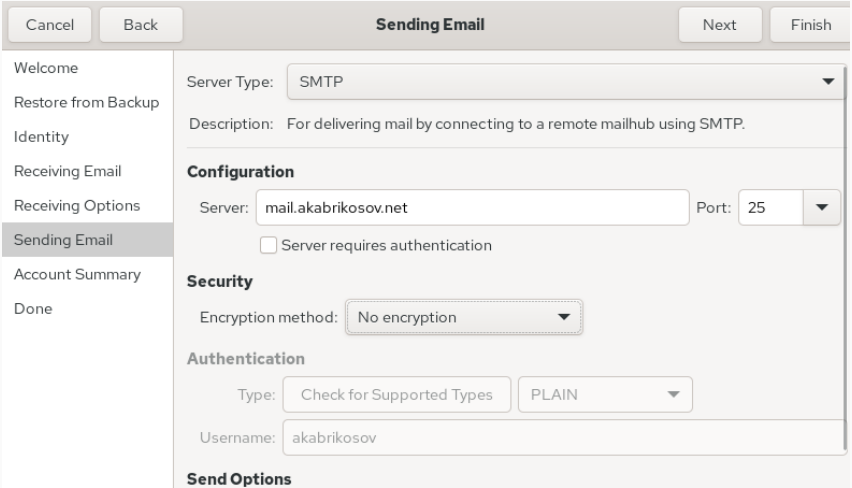


Рис. 9: Настройка входящей почты IMAP в Evolution

Evolution: исходящая почта (SMTP)

Проверены параметры SMTP: сервер mail.user.net, порт 25, без шифрования и без аутентификации.



Отправка и получение тестовых писем

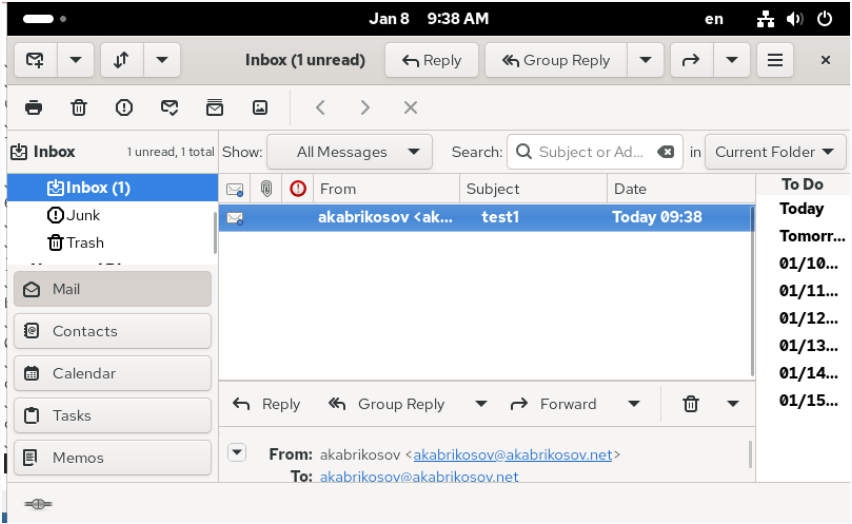


Рис. 11: Получение тестового письма в Evolution

Проверка POP3 через Telnet

```
1 701
2 701
3 701
.
retr 1
+OK 701 octets
Return-Path: <akabrikosov@akabrikosov.net>
X-Original-To: akabrikosov@akabrikosov.net
Delivered-To: akabrikosov@akabrikosov.net
Received: from client.akabrikosov.net (client.akabrikosov.net [192.168.1.30])
        by server.akabrikosov.net (Postfix) with ESMTP id 6A7BA412AD61
        for <akabrikosov@akabrikosov.net>; Thu,  8 Jan 2026 09:38:55 +0000 (UTC)
Message-ID: <4410500495ef26c431d47d8cfe57b95ae53b4d24.camel@akabrikosov.net>
Subject: test1
From: akabrikosov <akabrikosov@akabrikosov.net>
To: akabrikosov@akabrikosov.net
Date: Thu, 08 Jan 2026 09:38:55 +0000
Content-Type: text/plain
Content-Transfer-Encoding: 7bit
User-Agent: Evolution 3.52.4 (3.52.4-2.el10_1)
MIME-Version: 1.0

test1
.
dele 2
```

Выводы по проделанной работе

Вывод

В ходе лабораторной работы был установлен и настроен почтовый сервер на базе Postfix и Dovecot.

Выполнена настройка протоколов IMAP и POP3, обеспечено хранение почты в формате Maildir и корректная аутентификация пользователей.

Проверена работа почтовой системы с использованием Evolution, утилит mail/doveadm и проверки POP3 через Telnet.

Подтверждена корректная доставка, получение и удаление сообщений, а также готовность конфигурации к автоматическому развёртыванию в Vagrant.